

Current Position

University at Buffalo, SUNY
Department of Computer Science and Engineering
Assistant Professor

Education and Past Academic Positions

University of Maryland, College Park
Victor Basili Postdoctoral Research Fellow
Advised by Leonidas Lampropoulos

University of Pennsylvania
Ph.D. in Computer Science
Advised by Benjamin C. Pierce

Cornell University
M.Eng., B.S. in Computer Science
Advised by Adrian Sampson

Publications, Talks, and Grants

Dissertation

Property-Based Testing for the People
Harrison Goldstein *advised by* Benjamin C. Pierce
University of Pennsylvania, Defended May 24, 2024

Referreed Conference Publications

- PLDI'26 **The Search for Constrained Random Generators**
H. Goldstein, H. Peleg, C. Torczon, D. Sainati, L. Lampropoulos, B. C. Pierce
Programming Language Design and Implementation (PLDI) 2026. 25 pages.
- OOPSLA'26 **Fail Faster: Staging and Fast Randomness for High-Performance PBT**
Cynthia Richey, Joseph W. Cutler, **Harrison Goldstein**, Benjamin C. Pierce
Obj.-Oriented Programming, Sys., Langs., and Apps. (OOPSLA) 2026. 25 pages.
- ICFP'26
Conditional **Programmable Property-Based Testing**
Alperen Keles, Justine Frank, Ceren Mert, **Harrison Goldstein**, Leonidas Lampropoulos
International Conference on Functional Programming (ICFP) 2026. 25 pages.
- OOPSLA'25 **Tuning Random Generators: Property-Based Testing as Probabilistic Programming**
R. Tjoa, P. Garg, **H. Goldstein**, T. Millstein, B. C. Pierce, G. Van den Broek
Obj.-Oriented Programming, Sys., Langs., and Apps. (OOPSLA) 2025. 25 pages.
- OOPSLA'25 **QED in Context: An Observation Study of Proof Assistant Users**
Jessica Shi, Cassia Torczon, **Harrison Goldstein**, Benjamin C. Pierce, Andrew Head
Obj.-Oriented Programming, Sys., Langs., and Apps. (OOPSLA) 2025. 25 pages.
- UIST'25 **FEDT: Supporting experiment design and execution in HCI fabrication research**
V. Savage, **H. Goldstein**, N. Püsök, J. Ren, B. Dutt, C. Nandi, L. Oehlberg
User Interface Software and Technology (UIST) 2025. 13 pages.

- CCS'25 **A Qualitative Analysis of Fuzzer Usability and Challenges**
Y. Zhao, W. Guo, **H. Goldstein**, D. Votipka, K. R. Fulton, M. Mazurek
Conference on Computer and Communications Security (CCS) 2025. 13 pages.
- UIST'24 **Tyche: Making Sense of Property-Based Testing Effectiveness**
Harrison Goldstein, Jeffrey Tao, Zac Hatfield-Dodds, Benjamin C. Pierce, Andrew Head
User Interface Software and Technology (UIST) 2024. 13 pages.
- PLDI'24 **Stream Types**
J. W. Cutler, C. Watson, E. Nkurumeh, P. Hilliard, **H. Goldstein**, C. Stanford, B. C. Pierce
Programming Language Design and Impl. (PLDI) 2024. 24 pages.
- PLDI'24 **Daedalus: Safer Document Parsing**
I. Diatchki, M. Dodds, **H. Goldstein**, B. Harris, D. Holland, B. Razet, C. Schlesinger, S. Winwood
Programming Language Design and Implementation (PLDI) 2024. 24 pages.
- ICSE'24 **Property-Based Testing in Practice**
Harrison Goldstein, Joseph W. Cutler, Daniel Dickstein, Benjamin C. Pierce, Andrew Head
International Conference on Software Engineering (ICSE) 2024. 23 pages.
- ICFP'23 **Reflecting on Random Generation**
Harrison Goldstein, Samantha Frohlich, Meng Wang, Benjamin C. Pierce
International Conference on Functional Programming (ICFP) 2023. 34 pages.
- OOPSLA'22 **Parsing Randomness**
Harrison Goldstein, Benjamin C. Pierce
Obj.-Oriented Programming, Sys., Langs., and Apps. (OOPSLA) 2022. 25 pages.
- ESOP'21 **Do Judge a Test by its Cover: Combining Combinatorial and Property-Based Testing**
Harrison Goldstein, John Hughes, Leonidas Lampropoulos, Benjamin C. Pierce
European Symposium on Programming (ESOP) 2021. 27 pages.

Workshop Papers and Experience Reports

- PLATEAU'26 **Aporia: Asking Questions to Elicit User Decisions**
Saketh Kasibatla, Sorin Lerner, Hila Peleg, Benjamin Pierce, **Harrison Goldstein**
PLATEAU Workshop 2026. 10 pages.
- PLATEAU'26 **What's in a specification?: Exploring Student Conceptions of Formal Specifications**
Harrison Goldstein, Cameron Moy, Daniel Patterson
PLATEAU Workshop 2026. 4 pages.
- DBTest'26 **DIRT: Database-Integrated Random Testing**
Alperen Keles, Ethan Chou, **Harrison Goldstein**, Leonidas Lampropoulos
DBTest Workshop 2026. 7 pages.
- PLATEAU'25 **Designing Proof Deautomation in Rocq**
Jessica Shi, Cassia Torczon, **Harrison Goldstein**, Benjamin C. Pierce, Andrew Head
PLATEAU Workshop 2025. 12 pages.
- OCaml'24 **Mica: Automated Differential Testing for OCaml Modules**
Ernest Ng, **Harrison Goldstein**, Benjamin C. Pierce
OCaml Workshop 2024. 2 pages.
- ICFP'23 **Etna: An Evaluation Platform for Property-Based Testing (Experience Report)**
J. Shi, A. Kelles, **H. Goldstein**, B. C. Pierce, L. Lampropoulos
International Conference on Functional Programming (ICFP) 2023. 17 pages.

- HATRA'22 **Some Problems with Properties: A Study on Property-Based Testing in Industry**
Harrison Goldstein, Joseph W. Cutler, Adam Stein, Benjamin C. Pierce, Andrew Head
Human Aspects of Types and Reasoning Assistants (HATRA) 2022. 8 pages.
- SysML'18 **Programming Language Support for Natural Language Interaction**
Alex Renda, Harrison Goldstein, Sarah Bird, Chris Quirk, Adrian Sampson
Conference on Machine Learning and Systems (SysML) 2018. 3 pages.

Demos

- SCF'24 **Supporting Characterization Experiments in Fabrication Research**
V. Savage, N. Püsök, H. Goldstein, C. Nandi, J. Yi Ren and L. Oehlberg
Symposium on Computational Fabrication (SCF) 2024.
- UIST'23 **Tyche: In Situ Analysis of Random Testing Effectiveness**
Harrison Goldstein, Benjamin C. Pierce, Andrew Head
User Interface Software and Technology (UIST) 2023.

Talks and Podcasts

- Talk **Metaprogramming the Next Generation of Testing Tools**
Harrison Goldstein
Lean Together 2026
- Talk
Invited **Property-Based Testing for the People**
Harrison Goldstein
Cornell + UT Software Engineering Seminar
- Talk **The Best New Programming Language is a Proof Assistant**
Harrison Goldstein
DC Systems 006
- Podcast
Invited **Haskell Interlude #59**
Samantha Frohlich, Wouter Swierstra, Harrison Goldstein
Haskell Interlude Podcast, December 2024
- Talk
Invited **My PhD Compass: 6 Ways to Guide a PhD Towards Success**
Harrison Goldstein
PL Mentoring Workshop @ ICFP, September 2024
- Podcast
Invited **Disseminate #55**
Jack Waudby, Harrison Goldstein
Disseminate Podcast, June 2024
- Talk **Usable Property-Based Testing**
Harrison Goldstein
Amazon Web Services, June 2024
- Talk
Invited **Consider Collaboration**
Samantha Frohlich, Harrison Goldstein
PL Mentoring Workshop @ POPL, January 2024
- Talk
Invited **Advancing Property-Based Testing in Theory and Practice**
Harrison Goldstein
Microsoft Research, UC Berkeley, Galois, Inc., UCSC, University of Bristol
- Talk **Property-Based Testing in Practice**
Harrison Goldstein
Jane Street Programming Languages Colloquium, December 2023

Podcast
Invited **TheForkJoin #2**
Oliver Flatt, Rachit Nigam, **Harrison Goldstein**
TheForkJoin Podcast, July 2024

Talk **Some Problems with Properties**
Harrison Goldstein
NJPLS, October 2022

Talk **Reflecting on Random Generation**
Harrison Goldstein
NJPLS, May 2022

Posters

ICFP'21 **Ungenerators**
Harrison Goldstein
ICFP 2021

POPL'20 **Algebraic Combinatorial Testing**
Harrison Goldstein
POPL 2020

Misc.

Blog Post **Think Globally, Discuss PL Locally**
Harrison Goldstein, Michael Greenberg, Noam Zilberstein
SIGPLAN PL Perspectives Blog, September 2025

Draft **Delimited Continuations and Monads**
Harrison Goldstein
Unpublished PhD Milestone Draft, April 2021

SIGBOVIK'21 **Everybody Clap Your Hands: The Cha-Cha Slide is Turing Complete**
Harrison Goldstein
Conf. of the ACH Special Interest Group on Harry Q. Bovik (SIGBOVIK) 2021. 5 pages.

Blog Post **Conferences after COVID: An Early Career Perspective**
J. W. Cutler, **H. Goldstein**, A. K. Hirsch, J. Hong, C. Nandi
SIGPLAN PL Perspectives Blog, March 2021

Funded Grant Proposals

Amazon Gift: A Unified Framework for Automated Test Input Generation Using Lean
Spring 2026

Secured funding from AWS to fund work on the next generation of property-based testing automation. This work includes new proof foundations for PBT generators, as well as new techniques for automatically obtaining generators for real testing use-cases.

Team: Michael Hicks (co-PI), **Harrison Goldstein** (co-PI), Benjamin C. Pierce (co-PI)

NSF #2402449 SHF: Medium: Usable Property-Based Testing
2023-2024

Made significant contributions, in the form of both research project ideas and grant writing, to help secure a NSF Medium grant. This grant helped to fund my dissertation work, and it will continue to fund follow-on projects.

Team: Benjamin Pierce (co-PI), Andrew Head (co-PI), **Harrison Goldstein** (contributor)

Amazon Research Award: TYCHE: An IDE for Property-Based Testing

Spring 2023

Secured funding from AWS to fund my ongoing work on user interfaces for property-based testing. This award includes the opportunity to collaborate with contacts at AWS, including Michael Hicks, on interfaces that support real industrial workloads.

Team: **Harrison Goldstein** (primary author), Benjamin Pierce (co-PI), Andrew Head (co-PI)

Awards and Honors

John C. Reynolds Doctoral Dissertation Award 2025

ACM Special Interest Group on Programming Languages

Morris and Dorothy Rubinoff Award 2024

University of Pennsylvania, Department of Computer and Information Science

Victor Basili Postdoctoral Fellowship

University of Maryland, Computer Science Department

Certificate in Engineering Leadership

Cornell Engineering Leadership Certification Program

1500 Day Crossword Streak

New York Times Games

Teaching and Advising

Current PhD Students

Julia Montuori-Ahuja

Francille Zhuang

Cassia Torczon (Coadvised with Benjamin C. Pierce)

Graduate Advisees

Collaborator and Research Mentor, Segev Elazar Mittelman

Collaborator and Research Mentor, Caspar Popova

Collaborator and Research Mentor, Alperen Keles

Collaborator and Research Mentor, Pierce Darragh

Collaborator and Research Mentor, Ryan Tjoa

Collaborator and Research Mentor, Joseph W. Cutler

Collaborator and Research Mentor, Jessica Shi

Masters Co-Advisor, Ernest Ng

Courses

UB CSE 305 "Introduction to Programming Languages"
Undergraduate Course, Spring 2026

UB CSE 702 "Effective and Usable Formal Methods"
Graduate Seminar, Fall 2025

Penn CIS 552 "Advanced Programming"
Prof. Stephanie Weirich
Head TA 2020-2021

Penn CIS 810 "Writing and Speaking with Style"
Prof. Benjamin C. Pierce
Head TA 2021

Cornell CS 3110 "Functional Programming"
Prof. Nate Foster
Head TA 2017

Cornell Engineering Leadership
Profs. Erica Dawson and Werner Zorman
Head TA 2017

Cornell CS 2112 "Object-Oriented Design and Data Structures (Honors)"
Prof. Dexter Kozen
TA 2016

Service

Conference Organizing and Reviewing

Steering Committee Member, Upstate PL — Ongoing

PC Member, OOPSLA 2026

PC Member, PLDI 2025

PC Member, PLATEAU 2025

Steering Committee Member, NJPLS

Student Reviewer, PLATEAU 2024

Co-Organizer, Celebration for Benjamin Pierce's 60th Birthday

PC Member, HATRA 2023

Organizer, NJPLS 2023

Social Chair, POPL 2021

A/V Coordinator, ICFP 2021

Mentorship

Ph.D. Mentor, SIGPLAN-M — Ongoing

Mentor, Cornell COMPASS Program, Spring 2026

Speaker, PL Mentoring Workshop @ ICFP 2024

Ph.D. Mentor, Penn CIS Mentoring

Speaker, PL Mentoring Workshop @ POPL 2024

Research Mentor, REPL REU 2023

Research Mentor, DeepSpec REU

Misc.

Editor in Chief, PLClub Blog 2021-2024

Employment

University at Buffalo, SUNY

Assistant Professor 2025–

University of Maryland, College Park

Victor Basili Postdoctoral Fellow 2024-2025

Working with Prof. Leonidas Lampropoulos on topics related to usable property-based testing.

Galois, Inc.

Research Intern 2023

Worked on two research papers related to the SafeDocs DARPA program. Provided expertise on testing that helped situate the papers in the broader research context.

Amazon Web Services

Research Intern 2020

Worked on Zelkova, a tool for analyzing and proving properties about AWS access policies. Encoded logical constraints from access policies as SMT formulas, in order to infer policy implications.

Broadway Technology

Software Engineer and Software Engineering Intern 2017–2019

Built mission-critical internal tools for the company's financial personnel, in particular facilitating a transition to new financial tracking software. Designed and implemented data connectors, financial calculations, web interfaces, and more.

Susquehanna International Group, LLC

Technology Intern 2016

Helped to implement a safety system, protecting the firm from anomalous trading behaviors. Built an engine for executing business rules as monitors for live trading activities.

Last Second Beach, LLC

Chief Technology Officer 2015–2016

Helped to lead a small, early stage start-up, focused on providing one-price vacations. Built a demo iOS application, helping the company to win a \$25,000 grant as part of a Business Plan Competition.